# Chapter 16 Notes

## Section 1

### Theorem 2.7

Here is a proof of the classical part of this theorem. Consider some model modal $\mathcal{M}$ for which the hypotheses of the problem hold

Atomic case: In this case $X$ is atomic. Thus it must be that $A = X$ and $Y = B$, meaning, because we have assumed $A \equiv B$ is valid in $\mathcal{M}$, $X \equiv Y$ (which is the same formula) is also valid in $\mathcal{M}$.

Negation case: In this case $X$ is of the form $\sim\phi$. Then either $A = X$ or $A$ is a subformula of $\phi$. In the first case, obviously it follows that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, $Y$ must be of the form $\sim\psi$ (where $\phi \equiv \psi$ is valid over $\mathcal{M}$ by assumption). Then $\phi \supset \psi$ is valid over $\mathcal{M}$ (by assumption), meaning $\sim\psi \supset \sim\phi$ is valid over $\mathcal{M}$. Similarly, $\sim\phi \supset \sim\psi$ is valid over $\mathcal{M}$ (because $\psi \supset \phi$ is valid over $\mathcal{M}$). Thus $\sim\phi \equiv \sim\psi$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Conjunction case: In this case $X$ is of the form $\phi \wedge \psi$. Then either $A = X$ or $A$ is a subformula of either $\phi$ or $\psi$. In the first case, obviously it follwos that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, $Y$ must be of the form $\theta \wedge \gamma$, where both $\theta \equiv \phi$ and $\gamma \equiv \psi$ are true in $\mathcal{M}$. Then $(\phi \wedge \psi) \equiv (\theta \wedge \gamma)$, that is, $X \equiv Y$, being valid in $\mathcal{M}$ follows easily.

Existential case: In this case $X$ is of the form $(\exists x)\phi(x)$. Then either $A = X$ or $A$ is a subformula of $\phi(x)$. Again, the first case is obvious. In the second case, we show that (1) $(\exists x)\phi(x) \supset (\exists x)\psi(x)$ is valid over $\mathcal{M}$. So, for any world $p$ of the model, suppose that $(\exists x)\phi(x)$ is true at $p$. Then $\phi(a)$ is true at $p$ for some $a \in \mathcal{D}$. It is easy to see that this means $\psi(a)$ msut be true at $p$ (because $A \equiv B$ is assumed to be valid over $\mathcal{M}$). Thus $(\exists x)\psi(x)$ is true at $p$. Hence, because (1) is true at all possible worlds of $\mathcal{M}$, it follows that it is valid over $\mathcal{M}$. Similar reasoning shows that the converse of (1) is valid over $\mathcal{M}$. Hence $(\exists x)\phi(x) \equiv (\exists x)\psi(x)$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

### Exercise 2.7

For this exercise, "Classical logic" as a reason in the derivation stands for something derived due to the assumption that the axiom system we are working in contains the standard rules for classical logic.

(a)

| | | |
|---|---|---|
| 1. | $((X \wedge Y) \supset X)$ | (Classical logic) |
| 2. | $\Box((X \wedge Y) \supset X)$ | (1, Necessitation rule) |
| 3. | $\Box(X \wedge Y) \supset \Box X$ | (2, K axiom) |

(b) For this, note that part (a) can of course be altered to prove that $\Box(X \wedge Y) \supset \Box Y$.

| | | |
|---|---|---|
| 1. | $\Box(X \wedge Y) \supset \Box X$ | (Part (a)) |
| 2. | $\Box(X \wedge Y) \supset \Box Y$ | (Part (a)) |
| 3. | $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ | (1, 2, Classical logic) |

(c)

1. $((X \wedge Y) \supset (X \wedge Y))$ (Classical logic)
2. $\Box((X \wedge Y) \supset (X \wedge Y))$ (1, Necessitation rule)
3. $\Box(X \wedge Y) \supset \Box(X \wedge Y)$ (2, K axiom)
4. $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ (3, Part (b), Classical logic)